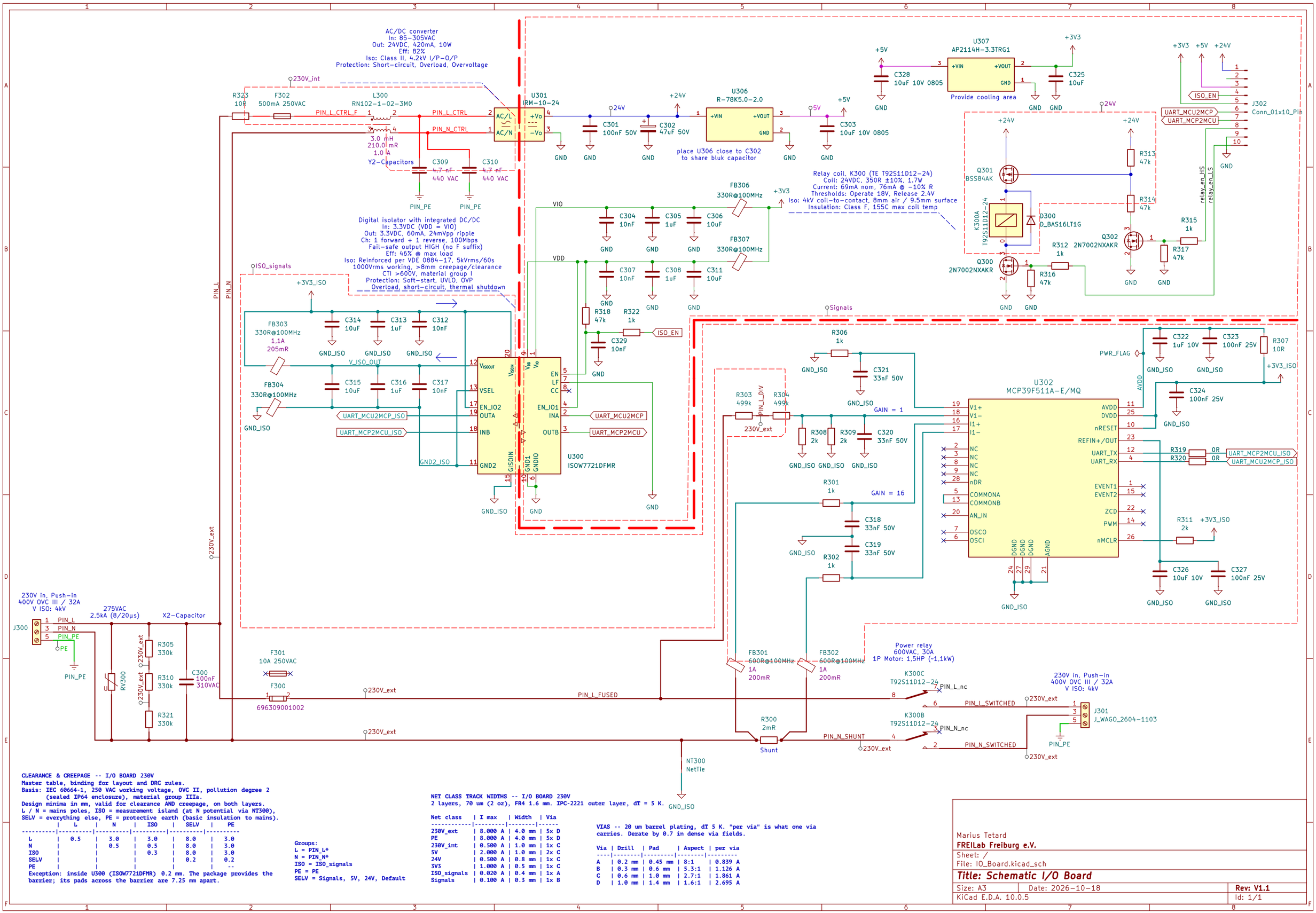

AC/DC converter
In: 85-305VAC
Out: 24VDC, 420mA, 10W
Eff: 82%
Iso: Class II, 4.2kV I/P-O/P
Protection: Short-circuit, Overload, Overvoltage
230V_int
R323 10R
F302 500mA 250VAC
L300 RN102-1-02-3M0
PIN_L_CTRL_F
PIN_L_CTRL
PIN_N_CTRL
3.0 mH
210.0 mR
1.0 A
Y2-Capacitors
C309 4.7 nF 440 VAC
C310 4.7 nF 440 VAC
PIN_PE
U301 IRM-10-24
AC/L
AC/N
+Vo
-Vo
24V
C301 100nF 50V
C302 47uF 50V
+24V
U306 R-78K5.0-2.0
+VIN
+VOUT
GND
5V
place U306 close to C302 to share bluk capacitor
C303 10uF 10V 0805
+5V
C328 10uF 10V 0805
U307 AP2114H-3.3TRG1
Provide cooling area
C325 10uF
+3V3
J302 Conn_01x10_Pin
ISO_EN
UART_MCU2MCP
UART_MCP2MCU
relay_en_HS
relay_en_LS
Digital isolator with integrated DC/DC
In: 3.3VDC (VDD = VIO)
Out: 3.3VDC, 60mA, 24mVpp ripple
Ch: 1 forward + 1 reverse, 100Mbps
Fail-safe output HIGH (no F suffix)
Eff: 46% @ max load
Iso: Reinforced per VDE 0884-17, 5kVrms/60s
1000Vrms working, >8mm creepage/clearance
CTI >600V, material group I
Protection: Soft-start, UVLO, OVP
Overload, short-circuit, thermal shutdown
VIO
VDD
FB306 330R@100MHz
FB307 330R@100MHz
C304 10nF
C305 1uF
C306 10uF
C307 10nF
C308 1uF
C311 10uF
R318 47k
R322 1k
C329 10nF
Relay coil, K300 (TE T92S11D12-24)
Coil: 24VDC, 350R ±10%, 1.7W
Current: 69mA nom. 76mA @ -10% R
Thresholds: Operate 18V, Release 2.4V
Iso: 4kV coil-to-contact, 8mm air / 9.5mm surface
Insulation: Class F, 155C max coil temp
Q301 BSS84AK
Q300 2N7002NXAKR
Q302 2N7002NXAKR
K300A T92S11D12-24
D300 D_BAS16LT1G
R312 1k
R313 47k
R314 47k
R315 1k
R316 47k
R317 47k
ISO_signals
+3V3_ISO
FB303 330R@100MHz 1,1A 205mR
FB304 330R@100MHz
C314 10uF
C313 1uF
C312 10nF
C315 10uF
C316 1uF
C317 10nF
GND_ISO
V_ISO_OUT
UART_MCU2MCP_ISO
UART_MCP2MCU_ISO
U300 ISOW7721DFMR
VISOOUT
VISOIN
VDD
VIO
EN
LF
CC
EN_IO2
OUTA
VSEL
INB
EN_IO1
INA
OUTB
GND2
GISOIN
GND1
GNDIO
Signals
R306 1k
C321 33nF 50V
GAIN = 1
R303 499k
R304 499k
230V_ext
R308 2k
R309 2k
C320 33nF 50V
R301 1k
GAIN = 16
C318 33nF 50V
C319 33nF 50V
R302 1k
U302 MCP39F511A-E/MQ
V1+
V1-
I1+
I1-
NC
nDR
COMMONA
COMMONB
AN_IN
OSCO
OSCI
AVDD
DVDD
nRESET
REFIN+/OUT
UART_TX
UART_RX
EVENT1
EVENT2
ZCD
PWM
nMCLR
DGND
AGND
C322 1uF 10V
C323 100nF 25V
R307 10R
PWR_FLAG
C324 100nF 25V
R319 0R
R320 0R
R311 2k
C326 10uF 10V
C327 100nF 25V
230V_ext
230V in, Push-in
400V OVC III / 32A
V ISO: 4kV
J300
PIN_L
PIN_N
PIN_PE
PE
275VAC
2,5kA (8/20µs)
X2-Capacitor
RV300
R305 330k
R310 330k
R321 330k
C300 100nF 310VAC
F301 10A 250VAC
F300 696309001002
PIN_L_FUSED
FB301 600R@100MHz 1A 200mR
FB302 600R@100MHz 1A 200mR
Power relay
600VAC, 30A
1P Motor: 1,5HP (~1,1kW)
K300C T92S11D12-24
K300B T92S11D12-24
PIN_L_nc
PIN_N_nc
PIN_L_SWITCHED
PIN_N_SWITCHED
J301 J_WAGO_2604-1103
R300 2mR
Shunt
PIN_N_SHUNT
NT300 NetTie
CLEARANCE & CREEPAGE -- I/O BOARD 230V
Master table, binding for layout and DRC rules.
Basis: IEC 60664-1, 250 VAC working voltage, OVC II, pollution degree 2
(sealed IP64 enclosure), material group IIIa.
Design minima in mm, valid for clearance AND creepage, on both layers.
L / N = mains poles, ISO = measurement island (at N potential via NT300),
SELV = everything else, PE = protective earth (basic insulation to mains).
L | 0.5 | 3.0 | 3.0 | 8.0 | 3.0
N | | 0.5 | 0.5 | 8.0 | 3.0
ISO | | | 0.3 | 8.0 | 3.0
SELV | | | | 0.2 | 0.2
PE | | | | | --
Exception: inside U300 (ISOW7721DFMR) 0.2 mm. The package provides the barrier; its pads across the barrier are 7.25 mm apart.
Groups:
L = PIN_L*
N = PIN_N*
ISO = ISO_signals
PE = PE
SELV = Signals, 5V, 24V, Default
NET CLASS TRACK WIDTHS -- I/O BOARD 230V
2 layers, 70 um (2 oz), FR4 1.6 mm. IPC-2221 outer layer, dT = 5 K.
Net class | I max | Width | Via
230V_ext | 8.000 A | 4.0 mm | 5x D
PE | 8.000 A | 4.0 mm | 5x D
230V_int | 0.500 A | 1.0 mm | 1x C
5V | 2.000 A | 1.0 mm | 2x C
24V | 0.500 A | 0.8 mm | 1x C
3V3 | 1.000 A | 0.5 mm | 1x C
ISO_signals | 0.020 A | 0.4 mm | 1x A
Signals | 0.100 A | 0.3 mm | 1x B
VIAS -- 20 um barrel plating, dT 5 K. "per via" is what one via carries. Derate by 0.7 in dense via fields.
Via | Drill | Pad | Aspect | per via
A | 0.2 mm | 0.45 mm | 8:1 | 0.839 A
B | 0.3 mm | 0.6 mm | 5.3:1 | 1.126 A
C | 0.6 mm | 1.0 mm | 2.7:1 | 1.861 A
D | 1.0 mm | 1.4 mm | 1.6:1 | 2.695 A
Marius Tetard
FREILab Freiburg e.V.
Sheet: /
File: IO_Board.kicad_sch
Title: Schematic I/O Board
Size: A3
Date: 2026-10-18
KiCad E.D.A. 10.0.5
Rev: V1.1
Id: 1/1